



- Salesforce has two ways to create an application—using point-and-click methods or by using the command line. The GUI method of application developing an application :

Create an Account :

- The first step is to create your Salesforce account. This is a fairly straightforward affair and it's something you've probably already done on dozens of web sites.
- You need to go to <http://developer.force.com/join>. Simply follow the steps (you'll need to answer name, password, username, and so forth) and you'll be set up.
- Once you log in to your Developer Edition organization, then log in to <https://login.salesforce.com>.

Create an Object :

- The next step is to create an object that will hold the data—our lunch expenses. Salesforce applications already contain a lot of prebuilt object called standard object for the data your applications need.
- However, you can build custom applications. Those are called custom objects.



1. Requirement Analysis

- Understand business needs and processes
- Identify users, data, security, and automation requirements
- Decide whether standard Salesforce features are sufficient or customization is needed

2. Choose Salesforce Cloud & Edition

- Select the appropriate cloud: **Sales Cloud, Service Cloud, Experience Cloud, Marketing Cloud**, etc.
- Choose the edition (Essentials, Professional, Enterprise, Unlimited) based on scalability and features

3. Org Setup

- Create a **Salesforce Org** (Developer / Sandbox / Production)
- Configure company details, time zone, language, and currencies

4. Data Modeling

- Define **Objects** (Standard or Custom)
- Create **Fields**, relationships (Lookup, Master–Detail)
- Design page layouts and record types

5. Security & Access Control

- Define **Profiles, Roles, and Permission Sets**
- Set **Object-level, Field-level, and Record-level** security
- Configure authentication and sharing rules



6. Customization & Configuration

- Configure **Validation Rules**
- Create **Workflows, Process Builder, or Flows**
- Customize UI using Lightning App Builder

7. Apex Development (if required)

- Write **Apex classes and triggers** for complex business logic
- Implement batch jobs, schedulers, and integrations

8. UI Development

- Build user interfaces using **Lightning Web Components (LWC)** or **Aura Components**
- Customize dashboards and reports

9. Integration

- Integrate Salesforce with external systems using
- REST / SOAP APIs
- Middleware (MuleSoft, Boomi, etc.)
- Implement real-time or batch integrations



10. Testing

- Perform unit testing and system testing
- Ensure **Apex test coverage $\geq 75\%$**
- Validate functionality, security, and performance

11. Deployment

- Use **Change Sets, Salesforce CLI, or DevOps tools**
- Deploy from Sandbox to Production

12. User Training & Documentation

- Train end users and administrators
- Prepare user guides and technical documentation

13. Maintenance & Support

- Monitor system performance
- Fix bugs, enhance features, and apply Salesforce updates



1. Requirement Analysis



2. System Design



3. Development



4. Testing



5. Deployment



6. Maintenance



CI/CD in Salesforce – Key Pointers :

- **CI (Continuous Integration)**
Automatically integrates code changes from multiple developers into a shared repository.
- **Version Control**
Uses Git (GitHub / GitLab / Bitbucket) to manage Salesforce metadata and source code.
- **Build Automation**
Salesforce CLI retrieves, converts, and validates metadata automatically.
- **Automated Testing**
Runs Apex unit tests to ensure $\geq 75\%$ **code coverage** and code quality.
- **CD (Continuous Deployment)**
Automatically deploys validated changes to Sandbox / Production.
- **Deployment Tools**
Uses Change Sets, Salesforce CLI, DevOps Center, or tools like Jenkins, GitHub Action



- To create the object, log in to your Salesforce account at <https://login.salesforce.com> and enter your username and password. create the custom object by following these steps:
- Select Setup in the upper-right corner of the page.
- Click Create Objects in the sidebar menu to show the Custom Objects page.
- Fill in the custom object definition. This is shown in Figure 11-5:
- Under Label, enter Lunch.
- Under Plural Label, enter Lunch.
- Under the Object Name, enter Lunch.
- Under the Description, enter Anobjectthatholdslunchexpenseinformation.
- Select the Allow Activities check box.
- Select the Allow Reports check box.
- Leave all the other values at their default levels.
- Click Save to finish the custom object creation.



The screenshot shows the Salesforce Developer Edition setup page in a web browser. The browser's address bar displays the URL: `https://na6.salesforce.com/ui/setup/Setup?setupid=DevTools&page=Setup&retUR`. The page header includes the Salesforce logo, navigation links (Setup, System Log, Help, Logout), and the text "force.com" and "Force.com".

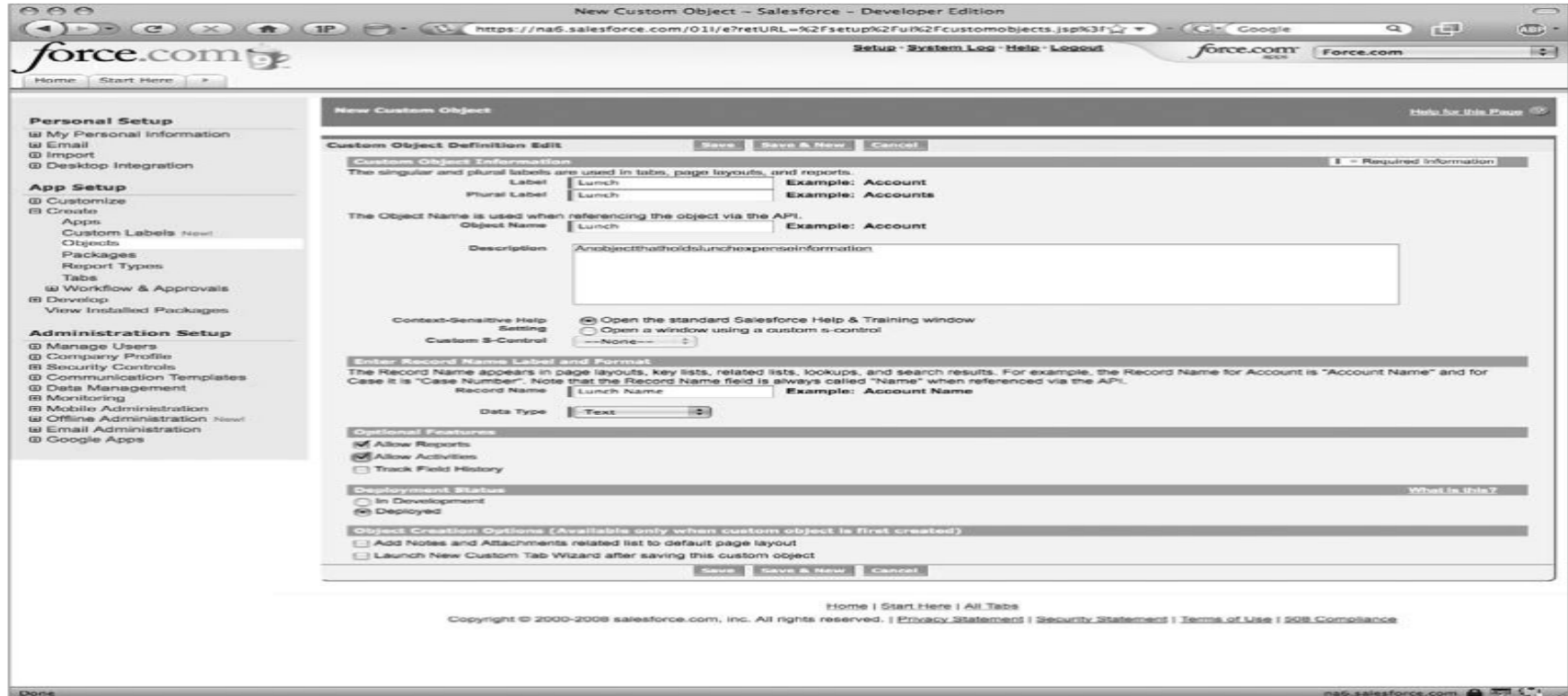
The main content area is divided into two columns. The left column contains a sidebar with the following sections:

- Personal Setup**
 - My Personal Information
 - Email
 - Import
 - Desktop Integration
- App Setup**
 - Customize
 - Create
 - Apps
 - Custom Labels New!
 - Objects
 - Packages
 - Report Types
 - Tabs
 - Workflow & Approvals
 - Develop
 - View Installed Packages
- Administration Setup**
 - Manage Users
 - Company Profile
 - Security Controls
 - Communication Templates
 - Data Management
 - Monitoring
 - Mobile Administration
 - Offline Administration New!
 - Email Administration
 - Google Apps

The right column contains the "Create" section, which is expanded to show the "Create" tab. This section lists the following items:

- Create**
 - Manage your custom apps
 - Manage Custom Labels and their translations for use in Visualforce pages and Apex code
 - Manage your custom objects
 - Share apps and components with other users via Force.com AppExchange
 - Manage your custom report types
 - Manage your custom tabs
 - Customize your Workflow Rules
 - Customize your Approval Processes
- Getting Started**
 - Building Custom Apps
 - Building Custom Objects, Tabs, and Related Lists
 - Publishing Apps on Force.com AppExchange
 - Tips & Hints for Workflow

The footer of the page includes the text "Home | Start Here | All Tabs" and "Copyright © 2000-2008 salesforce.com, Inc. All rights reserved. | Privacy Statement | Security Statement | Terms of Use | SOA Compliance".



New Custom Object – Salesforce – Developer Edition

https://na6.salesforce.com/011/e?retURL=%2Fsetup%2Fu%2Fcustomobjects.jsp%3F... Google

Setup - System Log - Help - Logout

force.com

Home Start Here

Personal Setup

- My Personal Information
- Email
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- Desktop Integration

App Setup

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Administration Setup

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New Custom Object [Help for this Page](#)

Custom Object Definition Edit [Save](#) [Save & New](#) [Cancel](#)

Custom Object Information [Required Information](#)

The singular and plural labels are used in tabs, page layouts, and reports.

Label Example: Account

Plural Label Example: Accounts

The Object Name is used when referencing the object via the API.

Object Name Example: Account

Description

Context-Sensitive Help Setting ☒ Open the standard Salesforce Help & Training window
☐ Open a window using a custom s-control
Custom S-Control

Enter Record Name, Label and Format

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Number". Note that the Record Name field is always called "Name" when referenced via the API.

Record Name Example: Account Name

Date Type

Optional Features

- ☒ Allow Reports
- ☒ Allow Activities
- ☐ Track Field History

Deployment Status [What is this?](#)

- ☐ In Development
- ☒ Deployed

Object Creation Options (Available only when custom object is first created)

- ☐ Add Notes and Attachments related list to default page layout
- ☐ Launch New Custom Tab Wizard after saving this custom object

[Save](#) [Save & New](#) [Cancel](#)

Home | Start Here | All Tabs

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Done na6.salesforce.com

FIGURE 11-5 Start creating your app by defining a custom object.



Fields :

- After the custom object is made to hold your lunch expenses, the next step is to add fields that capture the date of your business lunch, how much you spent, and the person you entertained.
- **Date First, create the Date field (shown in Figure 11-6):**
- Scroll down to the Custom Fields & Relationships related list.
- Select New to start the New Custom Field Wizard.
- Under Data Type, select Date and click Next
- **Enter the following data (shown in Figure 11-7):**
- Under Field Label, enter Date.
- Under Field Name, enter Date.
- Under the Description, enter Dateoflunch.
- Check the Required check box.
- For the Default Value, enter Today().
- Click Next, accept the defaults, and click Next again.
- Click Save & New to create the Date field and to restart the wizard for the next field

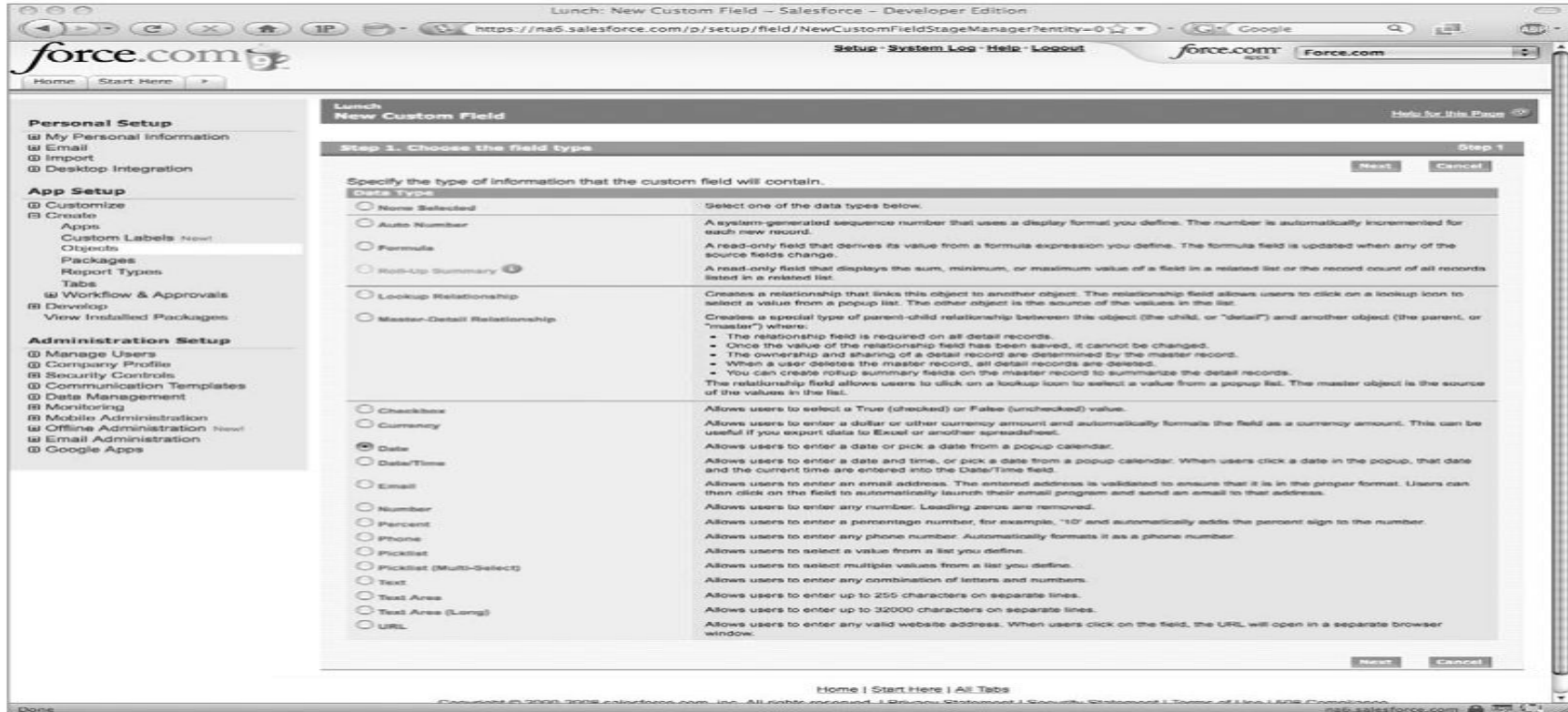


FIGURE 11-6 A key piece of data to be tracked by this app is the date.



Lunch: New Custom Field – Salesforce – Developer Edition

https://na6.salesforce.com/p/setup/field/NewCustomFieldStageManager

Setup · System Log · Help · Logout

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Lunch: New Custom Field

Help for this Page

Step 2: Enter the details Step 2 of 4

Previous Next Cancel

Field Label

Field Name

Description

Help Text

Required ☒ Always require a value in this field in order to save a record

Default Value [Show Formula Editor](#)

Use formula syntax, e.g., Text in double quotes: "hello", Number: 25, Percent as decimal: 0.10, Date expression: Today() + 7

Previous Next Cancel

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Done

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FIGURE 11-7 When filled, the values in the Date field should look like this.



- **Cost** The next field to generate is the Cost field. This is where the cost of your business lunch is stored. The wizard was automatically restarted in the previous section.
- For the Data Type, select Number (shown in Figure 11-8).
- Fill in the fields as follows (shown in Figure 11-9):
 1. Under Field Label, enter Cost.
 2. For the Length, enter 4.
 3. Under Decimal Places, enter 2.
 4. For the Field Name, enter Cost.
 5. For the Description, enter Costoflunch.
 6. Check the Required check box.
 7. Click Next, accept the defaults, and click Save & New to create the next field.

Lunch: New Custom Field – Salesforce – Developer Edition

https://na6.salesforce.com/p/setup/field/NewCustomFieldStageManagerEntity=0

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Lunch: New Custom Field

Help for this Page

Step 1. Choose the field type

Specify the type of information that the custom field will contain.

Next Cancel

Choose a type	Select one of the data types below.
☐ None Selected	
☐ Auto Number	A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.
☐ Formula	A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.
☐ Roll-Up Summary	A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.
☐ Lookup Relationship	Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.
☐ Master-Detail Relationship	Creates a special type of parent-child relationship between this object (the child, or "detail") and another object (the parent, or "master") where: - The relationship type is required on all detail records. - Once the value of the relationship field has been saved, it cannot be changed. - The ownership and sharing of a detail record are determined by the master record. - When a user deletes the master record, all detail records are deleted. - You can create rollup summary fields on the master record to summarize the detail records. The relationship field allows users to click on a lookup icon to select a value from a popup list. The master object is the source of the values in the list.
☐ Checkbox	Allows users to select a True (checked) or False (unchecked) value.
☐ Currency	Allows users to enter a dollar or other currency amount and automatically formats the field as a currency amount. This can be useful if you export data to Excel or another spreadsheet.
☐ Date	Allows users to enter a date or pick a date from a popup calendar.
☐ Date/Time	Allows users to enter a date and time, or pick a date from a popup calendar. When users click a date in the popup, that date and the current time are entered into the Date/Time field.
☐ Email	Allows users to enter an email address. The entered address is validated to ensure that it is in the proper format. Users can then click on the field to automatically launch their email program and send an email to that address.
☒ Number	Allows users to enter any number. Leading zeros are removed.
☐ Percent	Allows users to enter a percentage number, for example, "10" and automatically adds the percent sign to the number.
☐ Phone	Allows users to enter any phone number. Automatically formats it as a phone number.
☐ Picklist	Allows users to select a value from a list you define.
☐ Picklist (Multi-Select)	Allows users to select multiple values from a list you define.
☐ Text	Allows users to enter any combination of letters and numbers.
☐ Text Area	Allows users to enter up to 255 characters on separate lines.
☐ Text Area (Long)	Allows users to enter up to 32000 characters on separate lines.
☐ URL	Allows users to enter any valid website address. When users click on the field, the URL will open in a separate browser window.

Next Cancel

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FIGURE 11-8 The next field to define is the Cost field.



Lunch: New Custom Field -- Salesforce -- Developer Edition

https://na6.salesforce.com/p/setup/field/NewCustomFieldStageManager

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Lunch

New Custom Field

Help for this Page

Step 2. Enter the details

Step 2 of 4

Previous Next Cancel

Field Label

Please enter the length of the number and the number of decimal places. For example, a number with a length of 8 and 2 decimal places can accept values up to "12345678.90".

Length Decimal Places

Number of digits to the left of the decimal point

Number of digits to the right of the decimal point

Field Name

Description

Help Text

Required ☒ Always require a value in this field in order to save a record

Unique ☐ Do not allow duplicate values

External ID ☐ Set this field as the unique record identifier from an external system

Default Value

Use formula syntax, e.g., Text in double quotes: "hello", Number: 25, Percent as decimal: 0.15, Date expression: Today() + 7

Previous Next Cancel

Home | Start Here | All Tabs

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Done

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FIGURE 11-9 When filled, information in your Cost field will look like this.

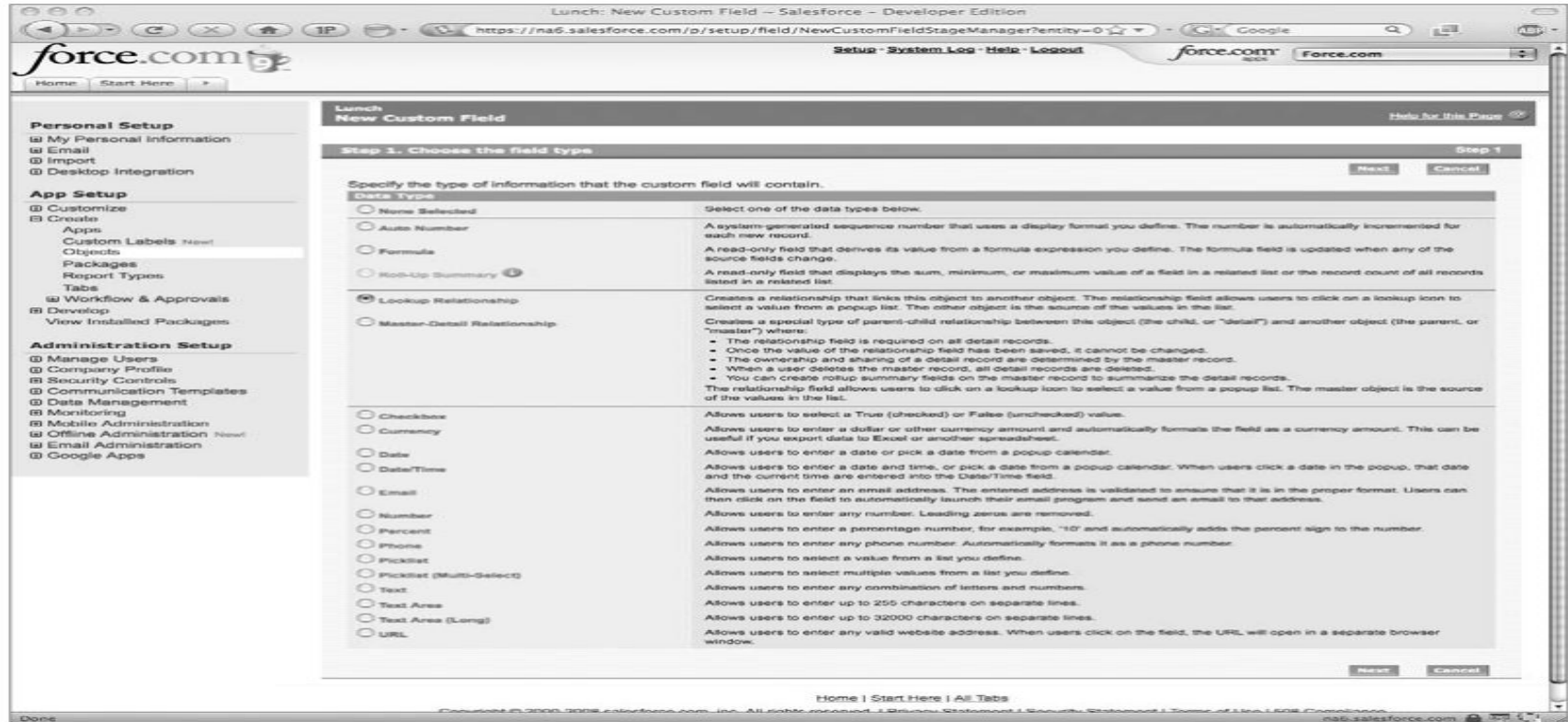


FIGURE 11-10 Selecting the Lookup Relationship data type allows you to link data for two objects.



- **Contact** The next field is the Contact field. This is the person with whom you had your business lunch. Again, if you clicked Save & New in the previous section, the field wizard restarts.
- For the Data Type, select Lookup Relationship (shown in Figure 11-10) and click Next.
- The Lookup Relationship data type allows you to link two data objects. In this example, we'll connect the Cost object and the Contact object.
- In the Related To drop-down list, select Contact and click Next.

Fill in the field details as follows (shown in Figure 11-11):

1. Under Field Label, enter Contact.
2. Under Field Name, enter Contact.
3. Under Description, enter PersonIhadlunchwith.
4. Leave the remaining defaults as they are, and then click Next.
5. Accept the defaults again and click Next.
6. Click Next two more times.
7. Click Save to finish up the field creation process.



Create a Tab :

- To show your application on the Salesforce web site, you need to create a tab to be added to the existing tabs. When users click this tab, they'll be able to track their lunch expenses.
1. On the startup page, click Create Tabs (this was the same screen where you created the object).
 2. Click New in the Custom Objects tab list to launch the New Custom Tab wizard.
 3. From the Object drop-down list, select Lunch.
 4. For the Tab Style, click the lookup icon and select the apple, shown in Figure 11-12 (it was our only food option).
 5. Accept the remaining defaults and click Next.
 6. Click Next and then Save to finish creating the tab.
 7. Once the tab has been created, it is added to your set of tabs

The image is a screenshot of a web browser displaying the Salesforce.com 'New Relationship' setup page. The browser's address bar shows the URL 'https://na6.salesforce.com/p/setup/field/NewCustomFieldStageManager'. The page title is 'Lunch: New Relationship -- Salesforce -- Developer Edition'. The Salesforce logo and navigation links like 'Setup', 'System Log', 'Help', and 'Logout' are visible at the top. On the left, a sidebar menu lists various setup categories: Personal Setup, App Setup, and Administration Setup. The main content area is titled 'Lunch New Relationship' and shows 'Step 3 of 6: Enter the label and name for lookup field'. It contains three input fields: 'Field Label' with the value 'Contact', 'Field Name' with the value 'Contact', and 'Description' with the value 'Personihadi lunchwith'. There are 'Previous', 'Next', and 'Cancel' buttons at the bottom of the form. The footer of the page includes copyright information for 2000-2006 and links to privacy and security statements.

FIGURE 11-11 The Contacts field should be configured to look like this.



Make the App

- At this point, we have created three fields and a tab. These are all pulled together to create our app. You make the app by following these steps:

1. Go to Setup Create Apps.
2. Click New to launch the New Custom App Wizard.

Enter the following details (shown in Figure 11-13):

1. Under App Label, enter Lunch Tracker.
2. Under the App Name, enter Lunch_Tracker
3. Under the Description, enter This application tracks your lunch expenses.
4. Click Next.
5. Accept the defaults and the default logo for the app, and then click Next.
6. In the Available Tabs box, locate the Lunch tab and click Add to add it to the selected tabs.
7. Leave the Default Landing tab set to the Home tab and click Next.
8. Select the Visible check box to make the app available to all users.
9. Click Save to create the Lunch Tracker app.
10. Repeat the steps 1–9, but this time select the Contacts tab. This is shown in Figure 11-14.
11. After the app is created, it is shown on the Force.com app menu in the upper right

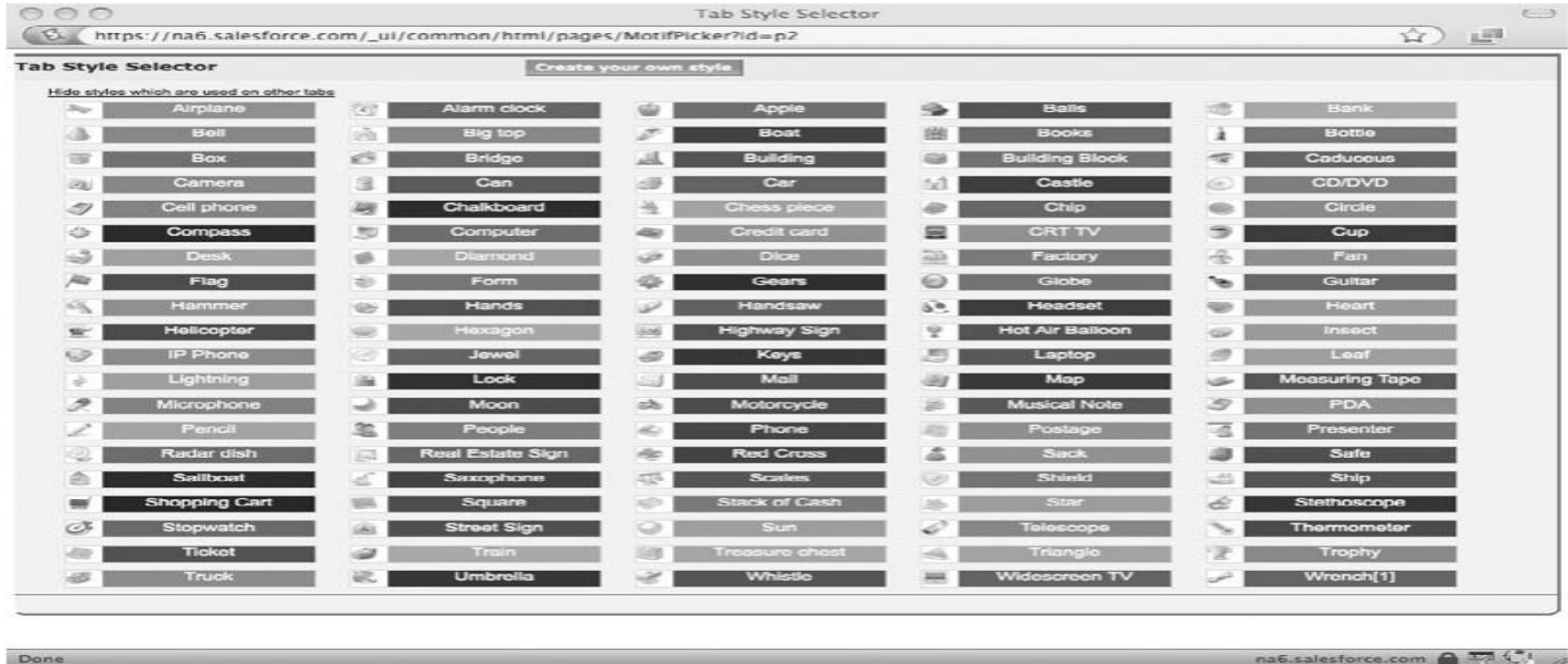


FIGURE 11-12 Start creating your app by defining a custom object.

The screenshot shows the 'New Custom App' wizard in the Salesforce Developer Edition. The browser address bar shows the URL: https://na6.salesforce.com/ui/setup/tab/TabSetWizard?retURL=%2F02u%3FretURL=. The page title is 'New Custom App - Salesforce - Developer Edition'. The left sidebar contains navigation links for 'Personal Setup' (My Personal Information, Email, Import, Desktop Integration), 'App Setup' (Customize, Create, Custom Labels, Objects, Packages, Report Types, Tabs, Workflow & Approvals, Develop, View Installed Packages), and 'Administration Setup' (Manage Users, Company Profile, Security Controls, Communication Templates, Data Management, Monitoring, Mobile Administration, Offline Administration, Email Administration, Google Apps). The main content area is titled 'New Custom App' and 'Step 1 of 4: Enter the Details'. It instructs the user to 'Fill in the fields below to define the custom app.' The 'Custom App Information' section includes fields for 'App Label' (Lunch Tracker), 'App Name' (Lunch_Tracker), and 'Description' (this application tracks your lunch expenses). A 'Next' button is visible at the bottom right of the form. The footer contains copyright information: Copyright © 2000-2006 salesforce.com, inc. All rights reserved. | Privacy Statement | Security Statement | Terms of Use | 508 Compliance.

FIGURE 11-13 Configure your app to have settings like this one.



Test It Out :

Now that your app has been created, let's take it for a spin.

1. Select the Lunch Tracker application from the Force.com app menu at the upper right of the screen.
 2. Click on the Lunch Tracker tab and click New to create a new lunch expense entry.
 3. Enter some test data (our example is shown in Figure 11-15).
 4. For the Contact field, click the lookup icon, click New to create a new contact named BruceDickinson, and then click Save.
 5. Click Save to save your record and to return to the detail page for the new record. And there you have it.
- You were able to make your very own app on Salesforce quickly and easily, and you didn't need to use any coding—it was all point and click.

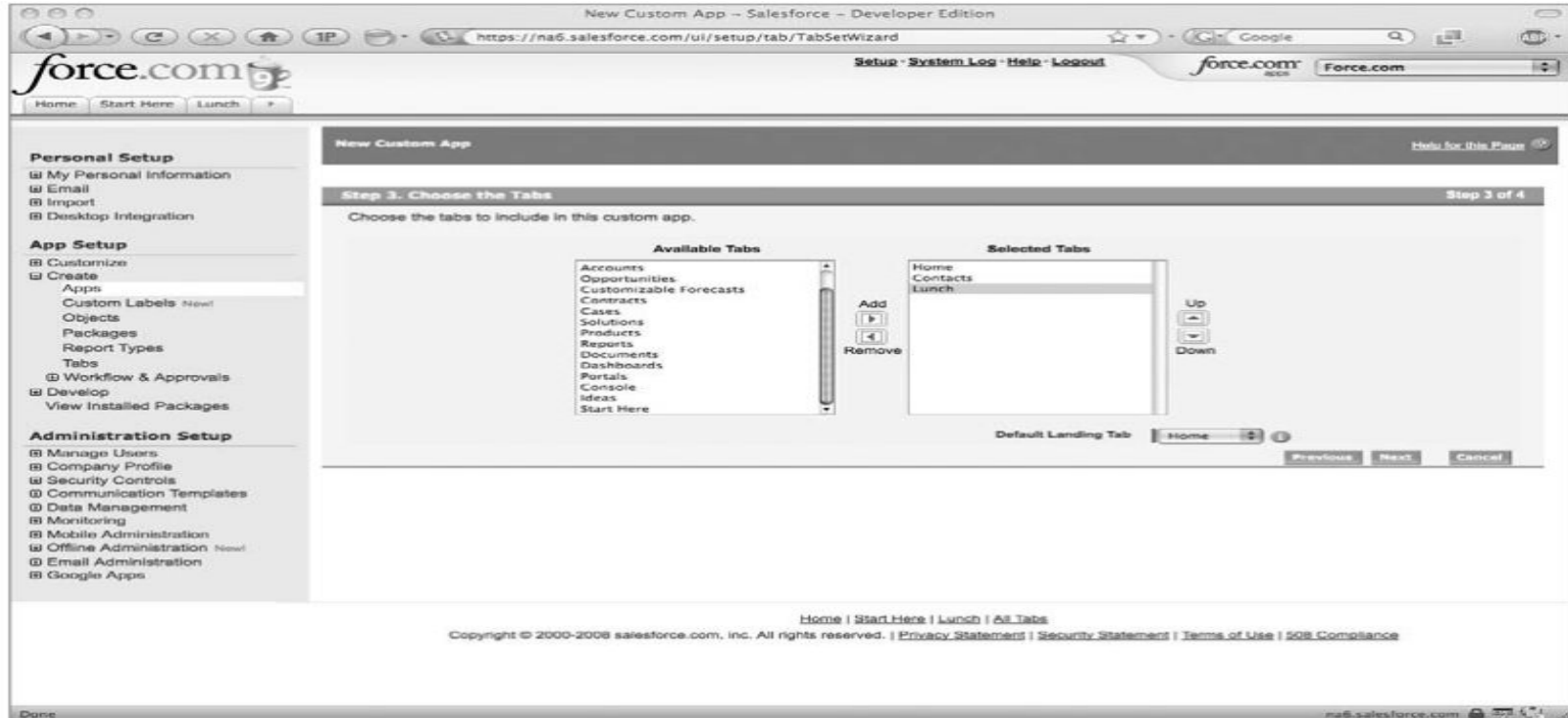


FIGURE 11-14 Choose Contacts and Lunch to be included in tabs.